Project Proposal

Title - Controlling the Weather: Time-based Dynamic Temperature in Story Generation

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Description - This project focuses on open-ended text generation, specifically story generation by large language models (LLMs). While recent models are capable of producing fluent text, generating longer stories that remain coherent, engaging, and narratively structured is still challenging.

Previous work has explored decoding strategies aimed at improving open-ended generation, such as nucleus sampling (Holtzman et al., 2019) and temperature optimization (Du et al., 2025). However, these approaches typically apply a fixed decoding strategy throughout the entire generation process. This contrasts with human writing, where different stages of a story (e.g., exposition, climax, resolution) are often approached with different levels of creativity and constraint.

Our central hypothesis is that varying the sampling temperature over time during generation can better reflect the structure of human storytelling, meaning it be more "human calibrated" and lead to improvements in perceived creativity, without substantially harming coherence.

To test this idea, we will use relatively small, locally runnable base models (e.g., Qwen2.5-0.5B or TinyLlama-1.1B or) and prompt them with examples from the *WritingPrompts* dataset (Fan et al., 2018). For each prompt, the model will generate a medium-length story (approximately 400–500 words), but the generation will be split into several consecutive segments. Each segment will be generated using **a different temperature value**, according to predefined strategies such as increasing temperature, decreasing (annealing) temperature, or mixed schedules.

Technically, this will be implemented by generating the story in chunks: the model first generates an initial sequence S_1 using temperature τ_1, then continues generation from the produced text using temperature τ_2, and so on, until the full story is produced.

We evaluate dynamic temperature schedules by comparing generated stories against a fixed-temperature baseline. Our primary evaluation uses the LLM-as-a-judge paradigm with a small open-source evaluator such as Prometheus-2, designed for fine-grained, instruction-following assessment (Kim et al., 2024; Pombal et al., 2025), or using Gemini models via API calls. When LLM-based evaluation is infeasible, we use non-LLM metrics: coherence, measured via average conditional log-likelihood of the text given its prefix (Su & Collier, 2023), and topic diversity, measured by n-gram overlap across generated stories, where lower overlap indicates higher diversity (Lee et al., 2022).

Requirements - The project will be conducted using locally available compute resources. We assume access to a single GPU with limited memory, which motivates the use of small (0.5-1B parameter) models and medium generation lengths (as the complexity is quadratic in sequence length). All models used will be open-weight and accessed via the Hugging Face Transformers library. The WritingPrompts dataset is publicly available and will be used only for prompting, not for training or fine-tuning. For evaluation via LLM, as stated, we will use Gemini API or on-prem Prometheus.

We would like to present our proposal in the final class 😊

Bibliography –

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Kim, S., Suk, J., Longpre, S., Lin, B. Y., Shin, J., Welleck, S., Neubig, G., Lee, M., Lee, K., & Seo, M. (2024). Prometheus 2: An Open Source Language Model Specialized in Evaluating Other Language Models. Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP).

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Su, Yixuan & Collier, Nigel. (2022). Contrastive Search Is What You Need For Neural Text Generation. 10.48550/arXiv.2210.14140.

AI reflection –

We used AI for rephrasing purposes, grammar checks, and APA styling. Also, we used it as an assistant when brainstorming solutions for evaluation methods.